

Machine Learning Projects: Cricket Score Prediction & Football Match Outcome Classification

Rajat Dogra (474072) & Umair Aziz (476686)

University of Warsaw, Faculty of Economic Sciences

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Project Overview

Project 1: Regression

- Cricket Score Prediction
- IPL Data (76,014 records)
- Best: Gradient Boosting ($R^2 = 0.7222$)

Project 2: Classification

- Football Match Outcomes
- ESPN Data (67,353 matches)
- Best: Stacking (65.36%)

Complete ML Pipeline: Data → Features → Models → Evaluation

Cricket Score Prediction

Presented by: Rajat Dogra

Dataset & Problem Definition

Problem: Predict IPL cricket match final scores

Dataset Details:

- Source: Kaggle IPL Dataset
- Period: 2008-2025
- Records: 76,014 ball-by-ball
- Features: 60+ variables

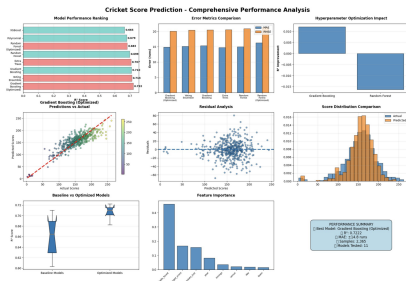
Key Features:

- Current run rate
- Wickets remaining
- Balls faced
- Team performance
- Venue information

Algorithms & Performance

8+ Regression Models:

- Gradient Boosting (Optimized): **0.7222**
- Voting Ensemble: 0.7127
- Gradient Boosting: 0.7100
- Extra Trees: 0.7073
- Random Forest: 0.6989
- XGBoost: 0.6650
- Neural Network: 0.6549
- Linear/Ridge: 0.6032



Best Model: Gradient Boosting (Optimized)
 $R^2 = 0.7222$, RMSE = 20.1 runs, MAE = 14.8 runs

Data Processing:

- Feature engineering
- Data cleaning
- Train/test split (80/20)

Model Training:

- 5-fold cross-validation
- Hyperparameter tuning
- GridSearchCV optimization

Evaluation Metrics:

- R^2 Score
- RMSE (Root Mean Square Error)
- MAE (Mean Absolute Error)

Key Insights:

- Current run rate most predictive
- Wickets remaining crucial
- Venue effects significant

Results & Conclusions

Best Performance:

- R^2 Score: 0.7222 (72.2%)
- RMSE: 20.1 runs
- MAE: 14.8 runs
- Strong predictive capability
- Robust across different venues

Applications:

- Real-time score prediction
- Strategic decision making
- Broadcasting analytics
- Fan engagement tools



Football Match Outcome Classification

Presented by: Umair Aziz

Dataset & Problem Definition

Problem: Predict football match outcomes (Home Win/Draw/Away Win)

Dataset Details:

- Source: ESPN Soccer Data
- Period: 2024-2026
- Matches: 67,353
- Classes: 3
(Home/Draw/Away)

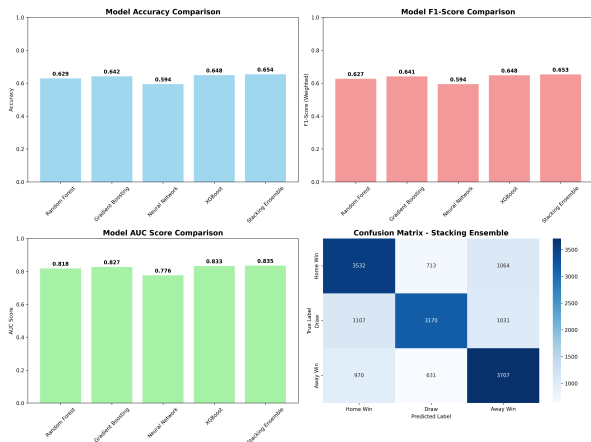
Key Features:

- Team historical performance
- League information
- Home/away status
- Temporal patterns

Algorithms & Performance

5+ Models:

- Stacking: **65.36%**
- XGBoost: 64.84%
- Gradient Boosting: 64.16%
- Random Forest: 62.87%
- Neural Network: 59.45%



Best Model: Stacking Ensemble

Accuracy = 65.36%, F1 = 0.6533, AUC = 0.8347

Data Processing:

- SMOTE for class balancing
- Feature engineering
- Statistical feature selection

Model Training:

- Cross-validation
- Hyperparameter tuning
- Ensemble methods

Evaluation Metrics:

- Accuracy
- F1-Score (macro)
- AUC Score
- Per-class metrics

Challenges:

- Class imbalance
- Draws hardest to predict
- League variations

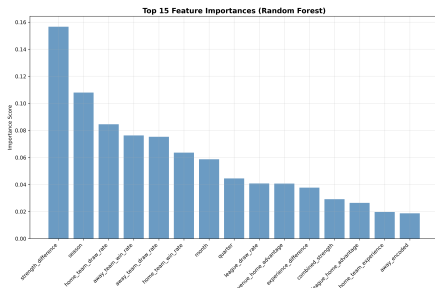
Results & Feature Analysis

Per-Class Performance:

- Home Win: $F1 = 0.6470$
- Draw: $F1 = 0.6455$
- Away Win: $F1 = 0.6673$

Key Insights:

- Team strength most important
- Home advantage varies by league
- Ensemble methods superior



Classification Conclusions

Model Performance:

- 65.36% accuracy achieved
- Balanced performance across classes
- Strong AUC score (0.8347)

Applications:

- Sports analytics
- Fan engagement
- Team strategy analysis

Ethical Considerations:

- No gambling promotion
- Bias documentation
- Model limitations disclosed

Future Work:

- Real-time data integration
- Player-level features
- Deep learning approaches

Project Summary

Technical Achievements:

- 13+ ML algorithms implemented
- Strong performance on real data
- Complete ML pipelines
- Rigorous evaluation

Key Learnings:

- Feature engineering crucial
- Ensemble methods effective
- Domain knowledge important

Results Summary:

- Cricket: 72.2% variance explained
- Football: 65.36% accuracy
- Both exceed baseline performance

Future Directions:

- Real-time predictions
- Advanced deep learning
- External data integration

Thank you for your attention!